

JAVA CLASSES ASSIGNMENT

Q1.

Code:

```
public class Student {  
    String name;  
    int rollno;  
    String phone;  
    String address;  
    public static void main(String[] args){  
        Student s1 = new Student();  
        s1.name = "John";  
        s1.rollno = 2;  
        s1.phone = "9876543210";  
        s1.address = "Mumbai";  
        Student s2 = new Student();  
        s2.name = "Sam";  
        s2.rollno = 3;  
        s2.phone = "0123456789";  
        s2.address = "Navi Mumbai";  
        System.out.println("Student 1 ");  
        System.out.println("Name: " +s1.name);  
        System.out.println("Rollno: " +s1.rollno);  
        System.out.println("Phone : " +s1.phone);  
        System.out.println("Address: " +s1.address);  
        System.out.println("Student 2 ");  
        System.out.println("Name: " +s2.name);  
        System.out.println("Rollno: " +s2.rollno);  
        System.out.println("Phone : " +s2.phone);  
        System.out.println("Address: " +s2.address);  
    }  
}
```

OUTPUT:

```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j
Student 1
Name: John
Rollno: 2
Phone : 9876543210
Address: Mumbai
Student 2
Name: Sam
Rollno: 3
Phone : 0123456789
Address: Navi Mumbai

Process finished with exit code 0
```

Q2.

Code:

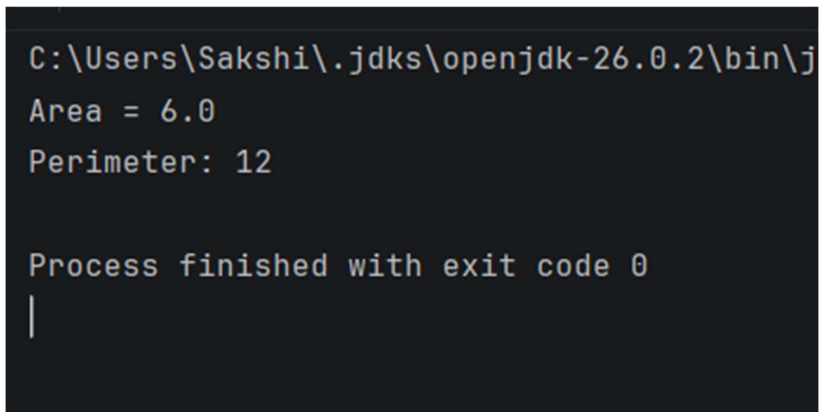
```
public class Triangle {
    int side1;
    int side2;
    int side3;
    Triangle(int a,int b, int c){
        side1 = a;
        side2 = b;
        side3 = c;
    }
    void area () {
        double s = (side1 + side2 + side3) / 2.0;
        double area = Math.sqrt(s * (s - side1) *
            (s - side2) * (s - side3));
    }
}
```

```

        System.out.println("Area = " +area);
    }
    void perimeter(){
        int perimeter = side1 + side2 + side3;
        System.out.println("Perimeter: " +perimeter);
    }
    public static void main(String[] args){
        Triangle t = new Triangle(3,4,5);
        t.area();
        t.perimeter();
    }
}

```

OUTPUT:



```

C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j
Area = 6.0
Perimeter: 12

Process finished with exit code 0
|

```

Q3.

CODE:

```

public class Rectangle {
    int length;
    int breadth;
    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }
}

```

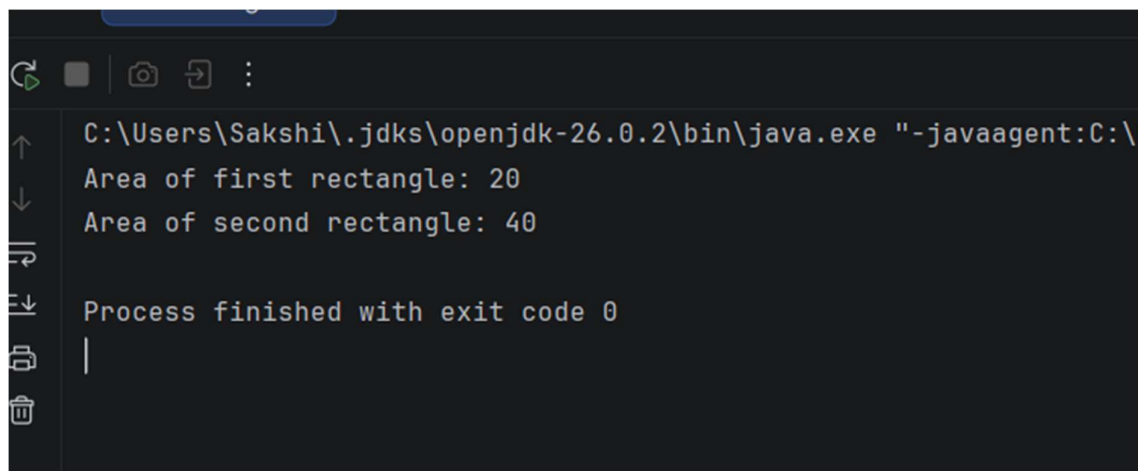
```

int area() {
    return length * breadth;
}

public static void main(String[] args) {
    Rectangle r1 = new Rectangle(4, 5);
    Rectangle r2 = new Rectangle(5, 8);
    System.out.println("Area of first rectangle: " + r1.area());
    System.out.println("Area of second rectangle: " + r2.area());
}
}

```

OUTPUT:



The screenshot shows a terminal window with the following output:

```

C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe "-javaagent:C:\
Area of first rectangle: 20
Area of second rectangle: 40

Process finished with exit code 0
|

```

Q4.

Code:

```

import java.util.Scanner;
class Complex {
    int real;
    int imaginary;
    Complex(int r, int i) {
        real = r;
        imaginary = i;
    }
}

```

```

    }
    void sum(Complex c) {
        int r = real + c.real;
        int i = imaginary + c.imaginary;
        System.out.println("Sum = " + r + " + " + i + "i");
    }
    void difference(Complex c) {
        int r = real - c.real;
        int i = imaginary - c.imaginary;
        System.out.println("Difference = " + r + " + " + i + "i");
    }
    void product(Complex c) {
        int r = (real * c.real) - (imaginary * c.imaginary);
        int i = (real * c.imaginary) +
            (imaginary * c.real);
        System.out.println("Product = " + r + " + " + i + "i");
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter real part of first number: ");
        int r1 = sc.nextInt();
        System.out.print("Enter imaginary part of first number: ");
        int i1 = sc.nextInt();
        System.out.print("Enter real part of second number: ");
        int r2 = sc.nextInt();
        System.out.print("Enter imaginary part of second number: ");
        int i2 = sc.nextInt();
        Complex c1 = new Complex(r1, i1);
        Complex c2 = new Complex(r2, i2);
        c1.sum(c2);
        c1.difference(c2);
        c1.product(c2);
    }
}

```

```
        sc.close();
    }
}
```

OUTPUT:



```
Run Complex x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe
Enter real part of first number: 3
Enter imaginary part of first number: 8
Enter real part of second number: 9
Enter imaginary part of second number: 14
Sum = 12 + 22i
Difference = -6 + -6i
Product = -85 + 114i

Process finished with exit code 0
```

Q5

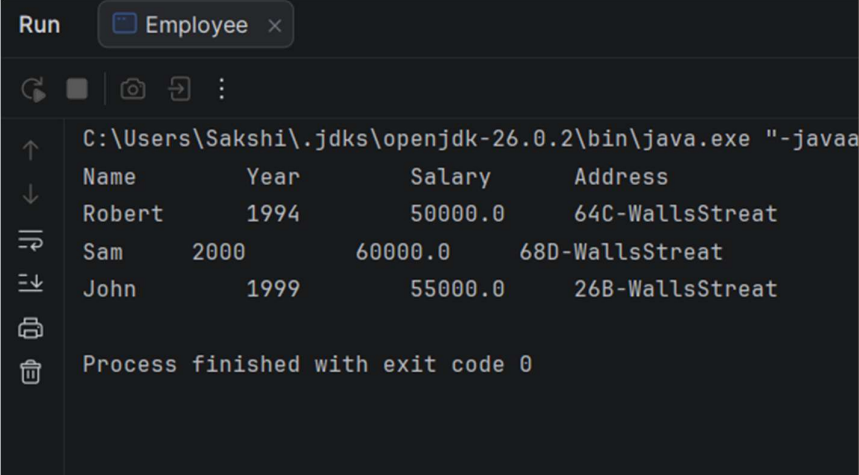
```
Code:public class Employee {
    String name;
    int year;
    double salary;
    String address;
    Employee(String n, int y, double s, String a) {
        name = n;
        year = y;
```

```

        salary = s;
        address = a;
    }
    void display() {
        System.out.println(name + "\t\t" + year + "\t\t" +
            salary + "\t\t" + address);
    }
    public static void main(String[] args) {
        Employee e1 = new Employee(
            "Robert", 1994, 50000, "64C-WallsStreat");
        Employee e2 = new Employee(
            "Sam", 2000, 60000, "68D-WallsStreat");
        Employee e3 = new Employee(
            "John", 1999, 55000, "26B-WallsStreat");
        System.out.println("Name\t\tYear\t\tSalary\t\tAddress");
        e1.display();
        e2.display();
        e3.display();
    }
}

```

Output:



```

Run Employee x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe "-javaa
Name      Year      Salary      Address
Robert    1994      50000.0     64C-WallsStreat
Sam       2000      60000.0     68D-WallsStreat
John      1999      55000.0     26B-WallsStreat
Process finished with exit code 0

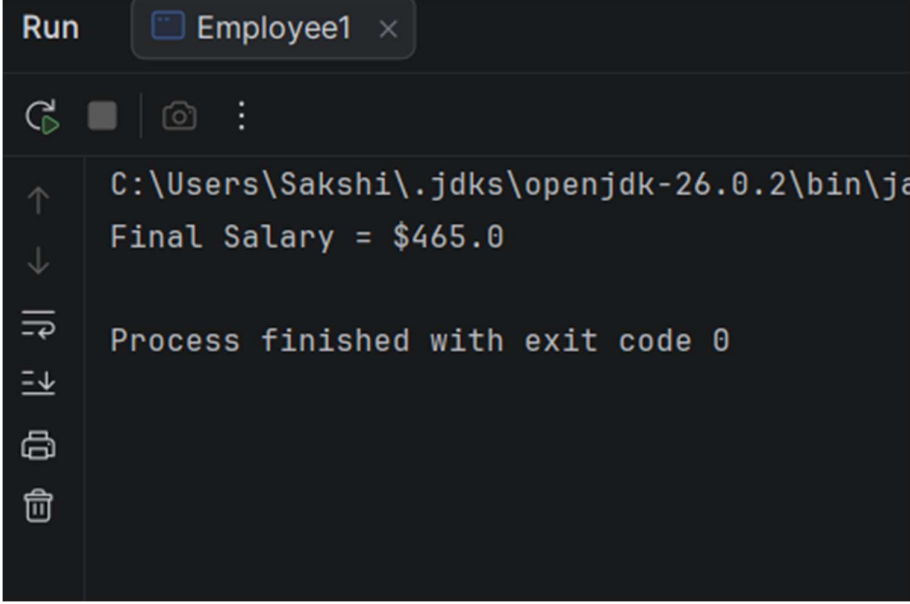
```

Q6.

CODE:

```
public class Employee1 {
    double salary;
    int hours;
    void getInfo(double s, int h) {
        salary = s;
        hours = h;
    }
    void addSal() {
        if (salary < 500) {
            salary = salary + 10;
        }
    }
    void addWork() {
        if (hours > 6) {
            salary = salary + 5;
        }
    }
    void displaySalary() {
        System.out.println("Final Salary = $" + salary);
    }
    public static void main(String[] args) {
        Employee1 e = new Employee1();
        e.getInfo(450, 8);
        e.addSal();
        e.addWork();
        e.displaySalary();
    }
}
```

OUTPUT:



The screenshot shows a dark-themed IDE window with a tab labeled "Employee1". Below the tab is a toolbar with icons for running, stopping, and debugging. The main area of the window displays the output of the program, which includes the Java command being executed and the resulting salary calculation.

```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java
Final Salary = $465.0

Process finished with exit code 0
```